distinctly Machian flavor, but it does not conform to the fully relationist perspective Mach advocated. Here's what I mean.

Mach argued[19] that when the spinning water's surface becomes concave, or when you feel your arms splay outward, or when the rope tied between the two rocks pulls taut, this has nothing to do with some hypothetical—and, in his view, thoroughly misguided—notion of absolute space (or absolute spacetime, in our more modern understanding). Instead, he argued that it's evidence of accelerated motion with respect to all the matter that's spread throughout the cosmos. Were there no matter, there'd be no notion of acceleration and none of the enumerated physical effects (concave water, splaying arms, rope pulling taut) would happen.

What does general relativity say?

According to general relativity, the benchmarks for all motion, and accelerated motion in particular, are freely falling observers—observers who have fully given in to gravity and are being acted on by no other forces. Now, a key point is that the gravitational force to which a freely falling observer acquiesces arises from all the matter (and energy) spread throughout the cosmos. The earth, the moon, the distant planets, stars, gas clouds, quasars, and galaxies all contribute to the gravitational field (in geometrical language, to the curvature of spacetime) right where you're now sitting. Things that are more massive and less distant exert a greater gravitational influence, but the gravitational field you feel represents the combined influence of the matter that's out there.[20] The path you'd take were you to give in to gravity fully and assume free-fall motion—the benchmark you'd become for judging whether some other object is accelerating —*would* be influenced by all matter in the cosmos, by the stars in the heavens and by the house next door. Thus, in general relativity, when an object is said to be accelerating, it means the object is accelerating with respect to a benchmark determined by matter spread throughout the universe. That's a conclusion which has the feel of what Mach advocated. So, in this sense, general relativity does incorporate some of Mach's thinking.

Nevertheless, general relativity does not confirm all of Mach's reasoning, as we can see directly by considering, once again, the